

# Bhavin K. Moriya

Data Scientist | Machine Learning | Privacy-Preserving Analytics

EU Blue Card Holder (Germany)

bhavinmoriya58@gmail.com

Google Scholar   GitHub   LinkedIn   Kaggle   YouTube   RoutingApp   StreamlitApps



## Experience

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| <b>Data Scientist / Research Assistant</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Hochschule Esslingen, Germany</b> | <b>Nov 2023 – Dec 2025</b> |
| <ul style="list-style-type: none"><li>• <b>AnoMoB Project:</b> Developed first-of-its-kind encrypted comparison operations and non-linear function evaluations using <b>OpenFHE (C++)</b> and <b>TFHE-rs (Rust)</b>.</li><li>• <b>Sustainability Analysis:</b> Leveraged <b>SUMO</b> to simulate movement data and conducted comparative analyses of travel modalities focusing on <b>CO2 emissions</b> and optimal route duration.</li><li>• <b>Decision Support for Sustainable Mobility:</b> Developed a <b>decision-support tool</b> for the ANOMOB project that uses a weighted scoring system to help users evaluate the trade-offs between travel time and CO2 emissions. It details a technical architecture based on Dockerized routing engines (ORS/OTP) and highlights research in privacy-preserving mobility analytics and homomorphic encryption.</li><li>• <b>Performance Benchmarking:</b> Co-authored breakthrough research comparing runtime and memory consumption of <b>CKKS</b> and <b>TFHE</b> schemes, published in Cryptology ePrint Archive (2025).</li><li>• <b>Data Engineering:</b> Built production-quality software tools for statistical analysis of mobility data using <b>Python</b>, <b>polars</b>, and <b>geopandas</b>.</li></ul> |                                      |                            |

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| <b>Assistant Professor</b>                                                                                                                                                                                                             | <b>Univ. of Viçosa, Brazil</b> | <b>Nov 2015 – Nov 2023</b> |
| <ul style="list-style-type: none"><li>• Taught undergraduate and master's level mathematics; supervised 3 dissertations.</li><li>• Collaborated with international researchers, resulting in 16+ peer-reviewed publications.</li></ul> |                                |                            |

### Postdoctoral Fellowships

- IMSc (2011–2013), HRI (2013–2014), CMI (2014), Univ. of Brasília (2014–2015)

## Education

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| • <b>Ph.D. in Mathematics</b> , Harish Chandra Research Institute<br>Thesis: Some Zero Sum Problems in Combinatorial Number Theory | <b>2011</b>        |
| • <b>M.Sc. &amp; B.Sc. in Mathematics</b> , The M.S. University of Baroda                                                          | <b>2003 – 2005</b> |

## Skills

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**Programming:** Python (Proficient), C++, Rust, SQL, ~~LaTeX~~  $\text{\LaTeX}$ , Go, Julia

**Data & Simulation:** polars, geopandas, SUMO, Scikit-learn, Docker

**Workflow:** LLM-assisted development (**GitHub Copilot**), Git, **Agile Research**, **Streamlit**

**Languages:** English (Fluent), German (**B1/Professional Communication**), Portuguese (Fluent/Professional), Hindi (Native), Gujarati (Native)

## Selected Projects (GitHub)

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- **Vascular Surrogate Models** – Accelerating Intracranial Aneurysm CFD.
- **AudioMNIST** – A spoken digit recognition system.

## Publications & Awards

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- (2025) **A Performance Comparison of Homomorphic Encryption Schemes CKKS and TFHE**, Cryptology ePrint Archive.
- 16 peer-reviewed publications in combinatorics and cryptography.
- Recipient of multiple international grants (ICM, CIMPA-ICTP, FAPEMIG).